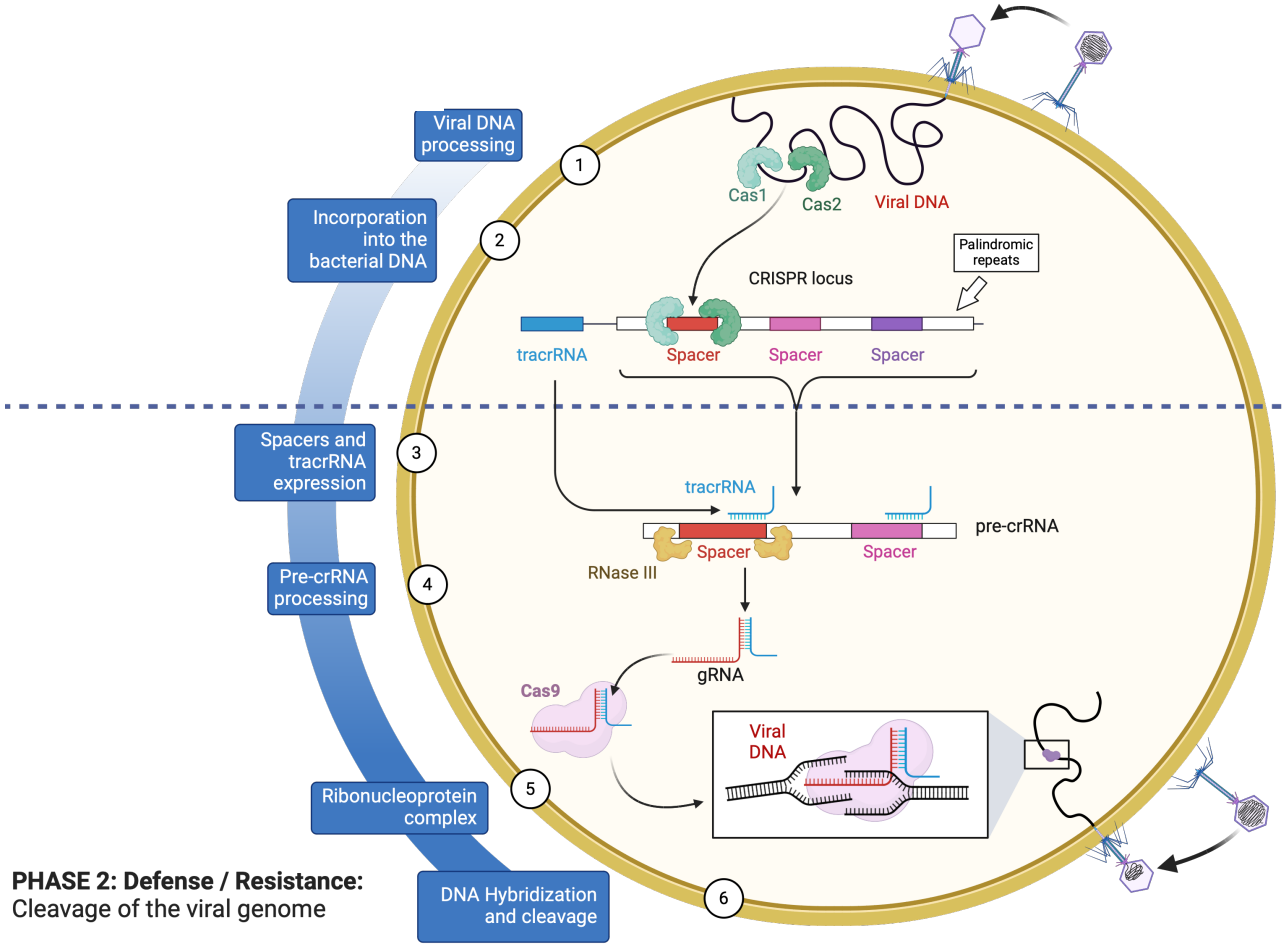
Genetic edition by CRISPR

TBL 7.2

OCTOBER 22, 2026

Dr. Serrano | dserrano@unav.es

WHAT IS CRISPR/Cas9?



PHASE 2: Defense / Resistance:
Cleavage of the viral genome

KEY COMPONENTS

CRISPR sequences. Clustered Regularly Interspaced Short Palindromic Repeats

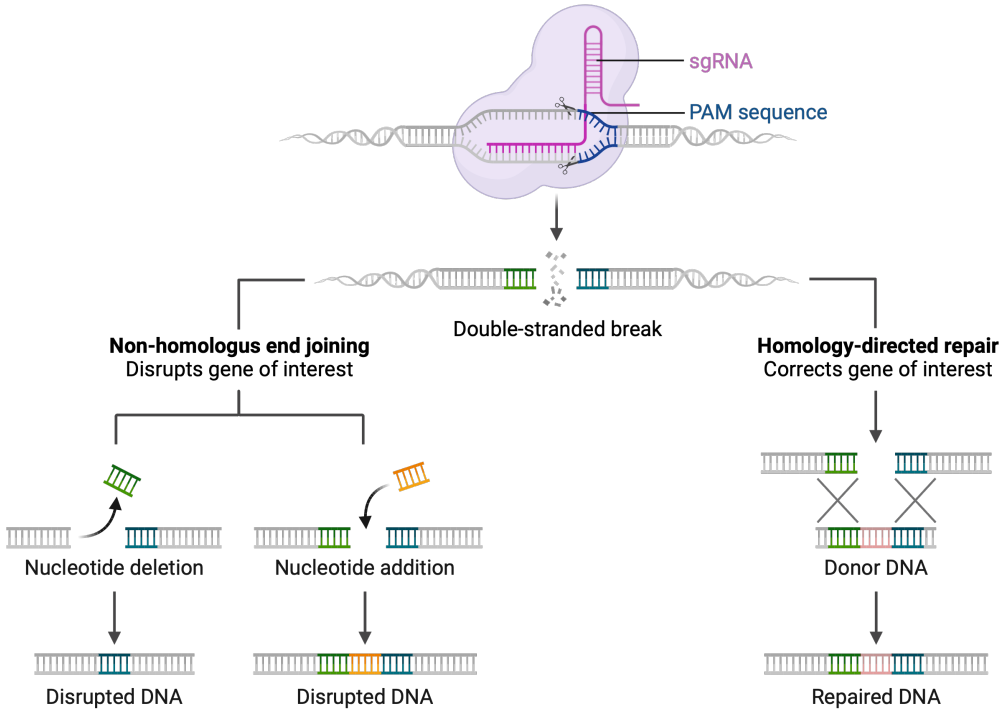
Spacers. DNA sequences derived from viruses that attacked the bacterium

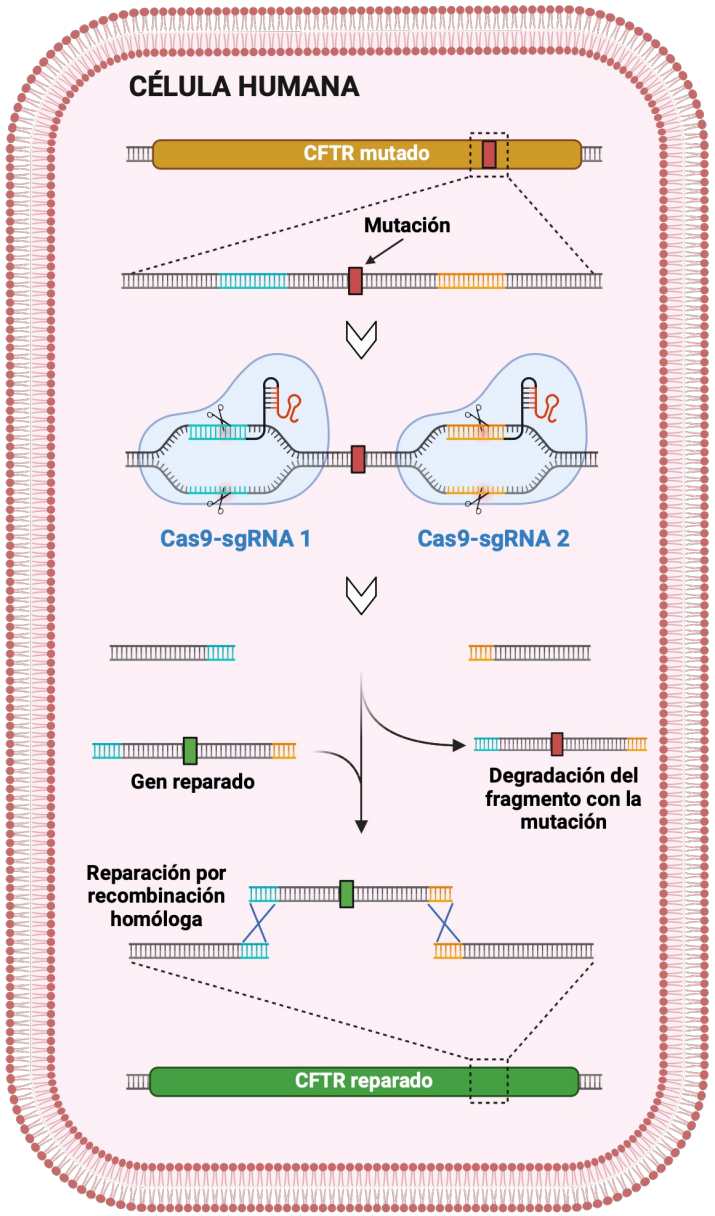
Cas9 protein. An enzyme that cuts DNA at specific locations

Guide RNA (gRNA, crRNA). Small RNA sequence matching a target DNA sequence, directing Cas9 to cut the DNA

tracrRNA (trans activating cripsr RNA) small scaffolding RNA that facilitates binding of gRNA to Cas9

CRISPR/Cas9 in research and Medicine





CRISPR/Cas9 in research and Medicine

Nobel prize in chemistry 2020

Jennifer Doudna and Emmanuel Charpentier



First CRISPR therapy - Casgevy

casgevy®
(exagamglogene autotemcel)
suspension for IV infusion

FDA Approves First Gene Therapies to Treat Patients with Sickle Cell Disease

December 08, 2023

FDA News Release

For Immediate Release:

Today, the U.S. Food and Drug Administration approved two milestone treatments, Casgevy and Lyfgenia, representing the first cell-based gene therapies for the treatment of sickle cell disease (SCD) in patients 12 years and older. Additionally, one of these therapies, Casgevy, is the first FDA-approved treatment to utilize a type of novel genome editing technology, signaling an innovative advancement in the field of gene therapy.

Sickle cell disease is a group of inherited blood disorders affecting approximately 100,000 people in the U.S. It is most common in African Americans and, while less prevalent, also affects Hispanic Americans. The primary problem in sickle cell disease is a mutation in hemoglobin, a protein found in red blood cells that delivers oxygen to the body's tissues. This mutation causes red blood cells to develop a crescent or "sickle" shape. These sickled red blood cells restrict the flow in blood vessels and limit oxygen delivery to the body's tissues, leading to severe pain and organ damage called vaso-occlusive events (VOEs) or vaso-occlusive crises (VOCs). The recurrence of these events or crises can lead to life-threatening disabilities and/or early death.

EUROPEAN MEDICINES AGENCY
SCIENCE MEDICINES HEALTH

14 December 2023
EMA/CHMP/506425/2023
Committee for Medicinal Products for Human Use (CHMP)

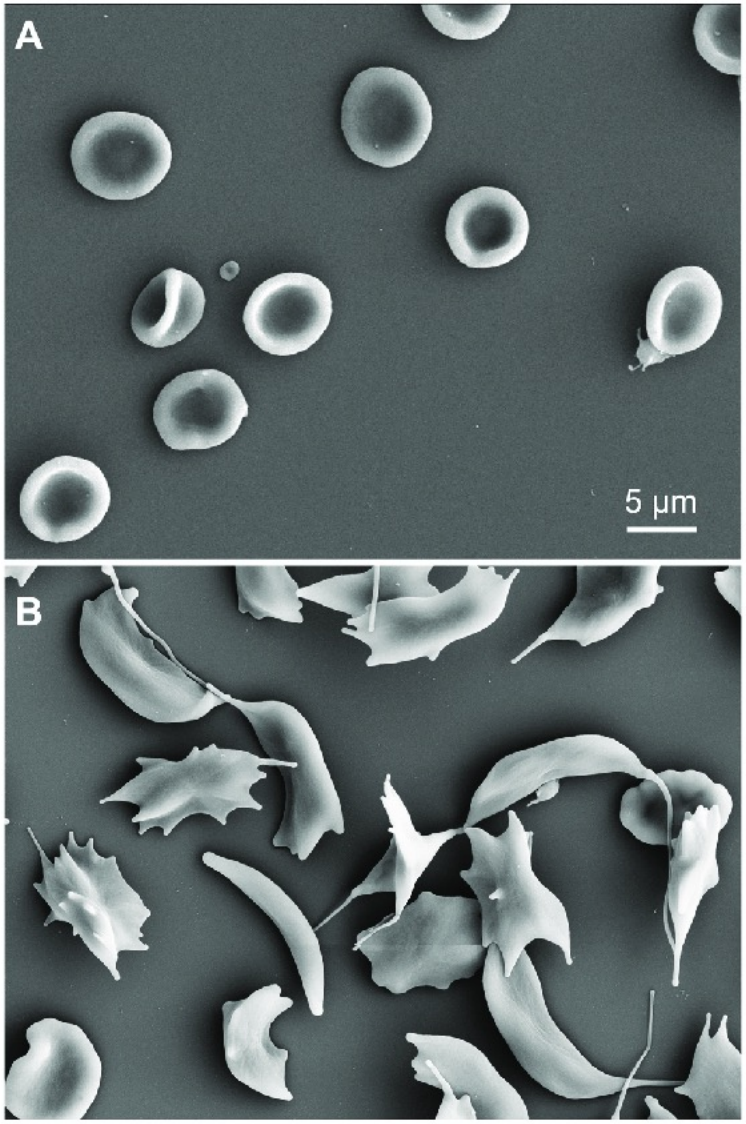
Summary of opinion[1] (initial authorisation)

Casgevy

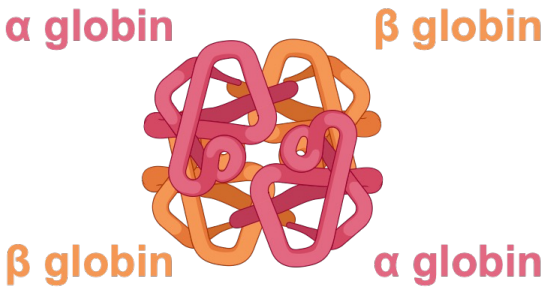
exagamglogene autotemcel

On 14 December 2023, the Committee for Medicinal Products for Human Use (CHMP) adopted a positive opinion, recommending the granting of a conditional[2] marketing authorisation for the medicinal product Casgevy[3], intended for the treatment of transfusion-dependent β-thalassemia (TDT) and sickle cell disease (SCD). As Casgevy is an advanced therapy medicinal product, the CHMP positive opinion is based on an assessment by the Committee for Advanced Therapies.

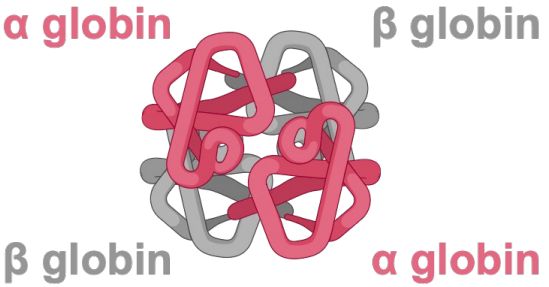
Thalassemia / Sickle cell disease



Normal hemoglobin



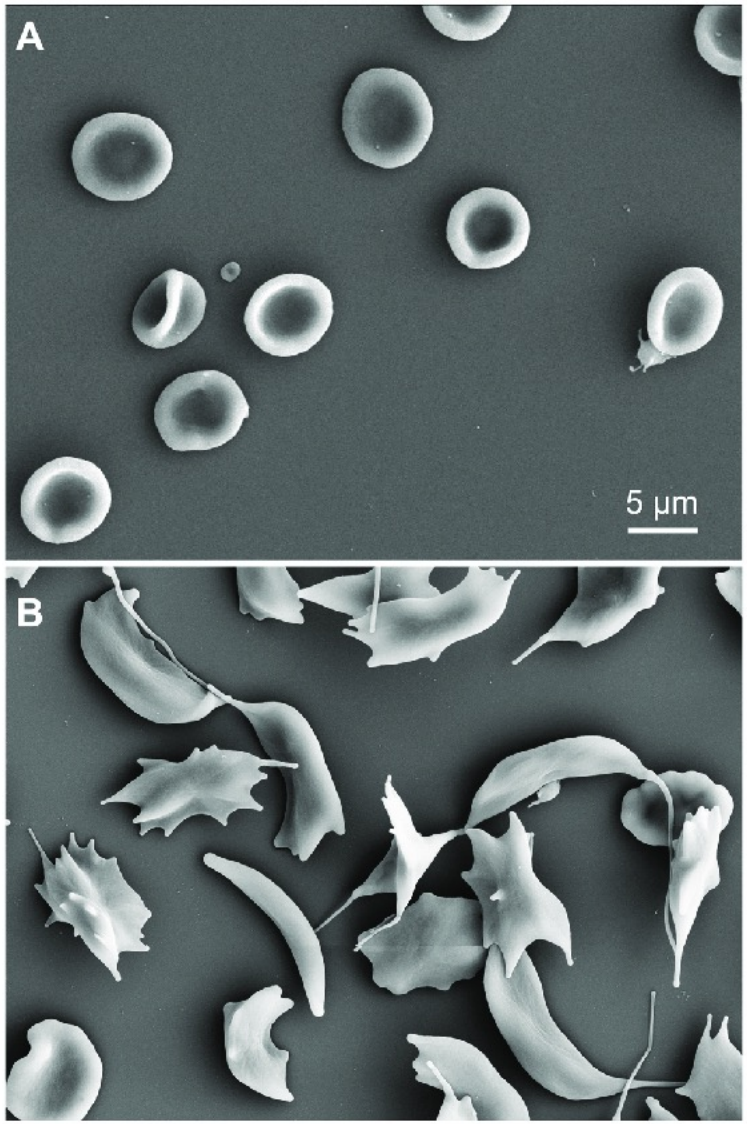
Mutant hemoglobin

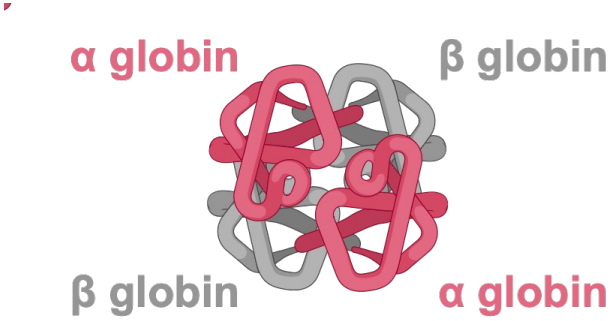


Beta thalassemia

Alpha thalassemia

Beta thalassemia





One gene affected => β thalassemia minor
Both genes affected => β thalassemia major **Severe**

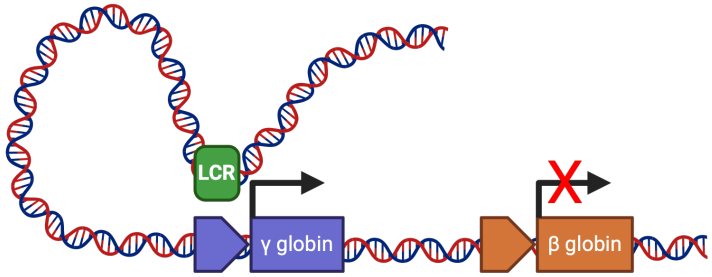
Symptoms: similar to anemia
- Feeling tired or weak
- Shortness of breath
- Paleness
- Dizziness and fainting
- Headaches

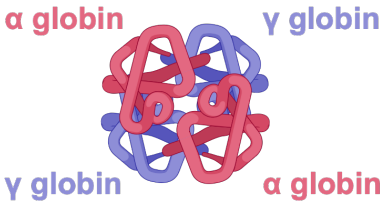
Complications (major thalassemia)
- Blood clots (coagules)
- Heart attack
- Liver and kidney failure
- Dementia
- Diabetes

Treatment (major thalassemia)
- Regular blood transfusions
- Iron chelation (caused by transf.)
- Bone marrow transplant

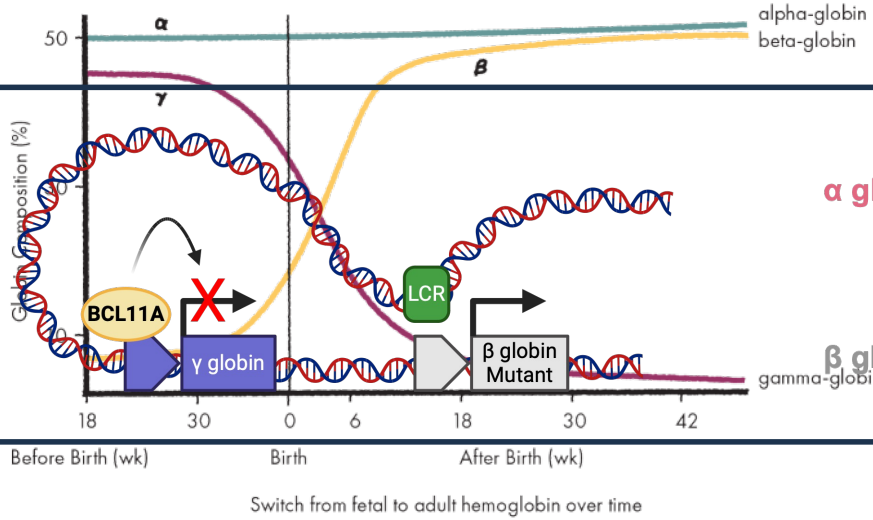
Gamma/beta globin gene regulation

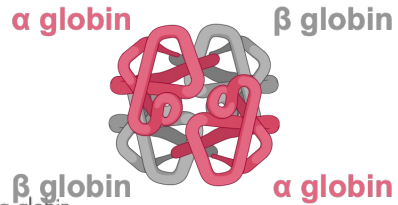
Fetal hemoglobin



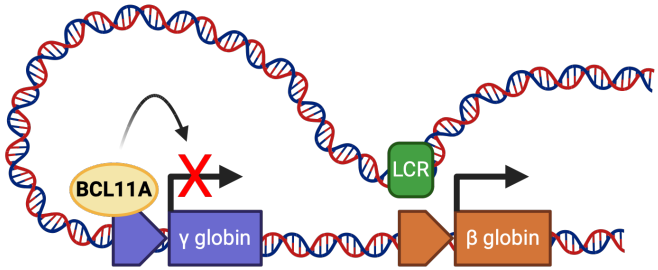


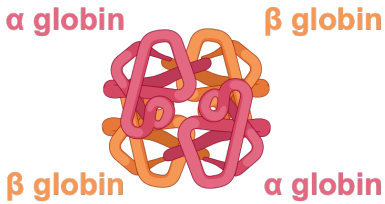
Beta thalassemia





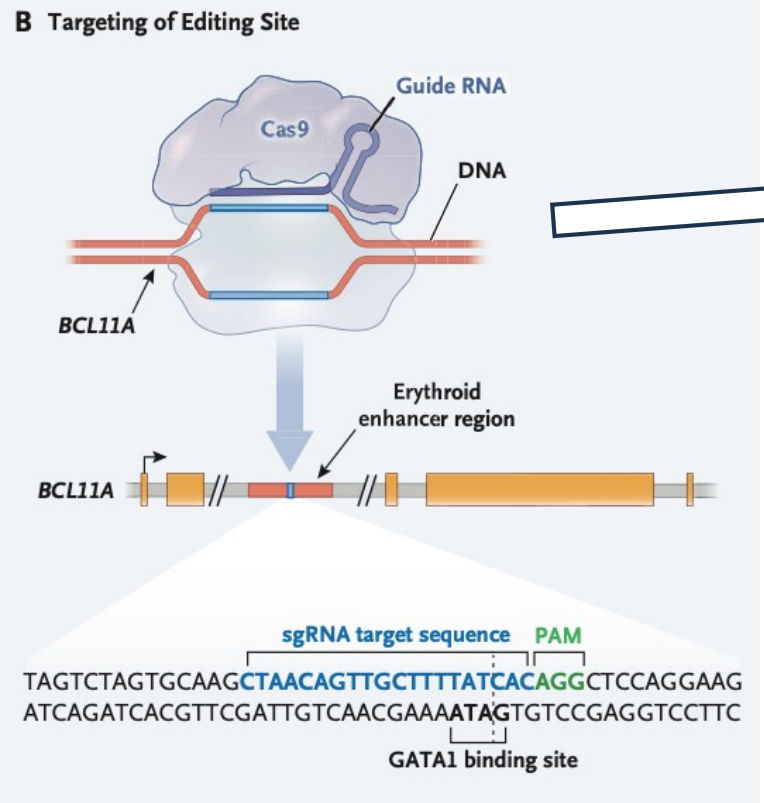
Adult hemoglobin



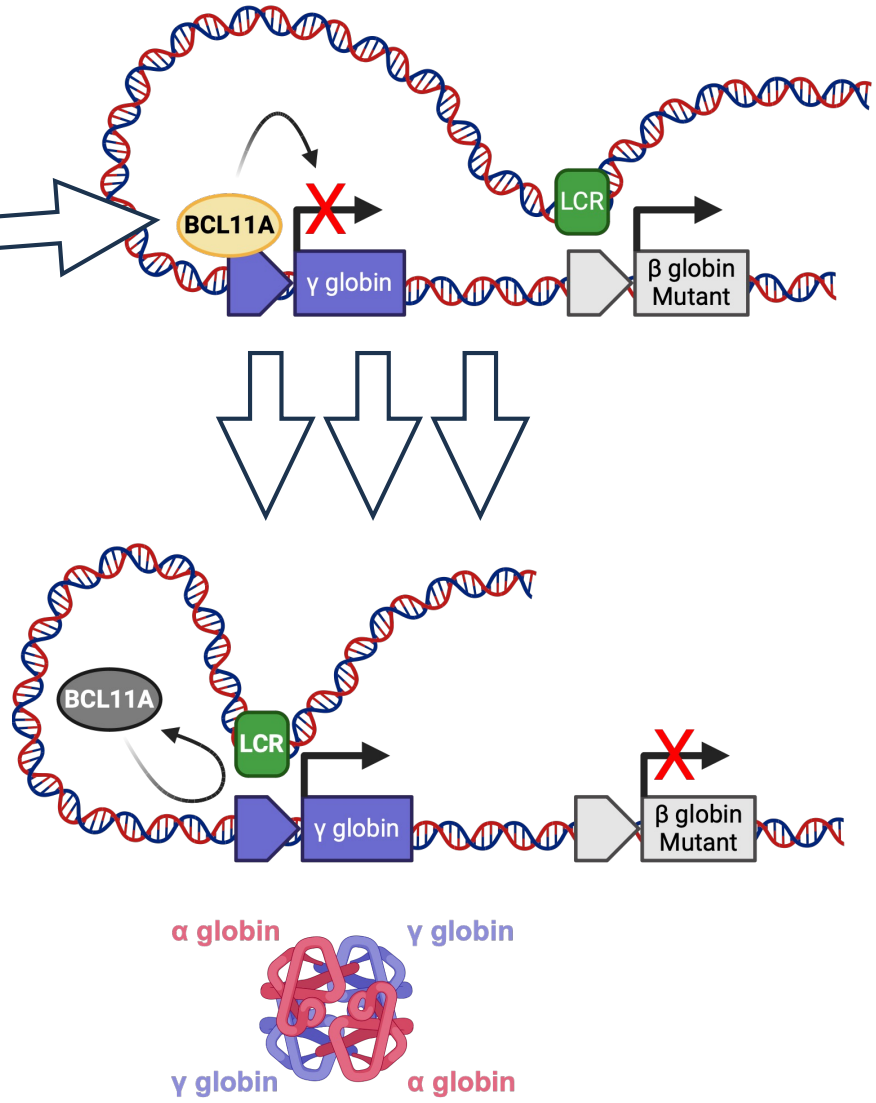


Casgevy – beta thalassemia / SCD treatment

CRISPR/cas9 modification of BCL11A

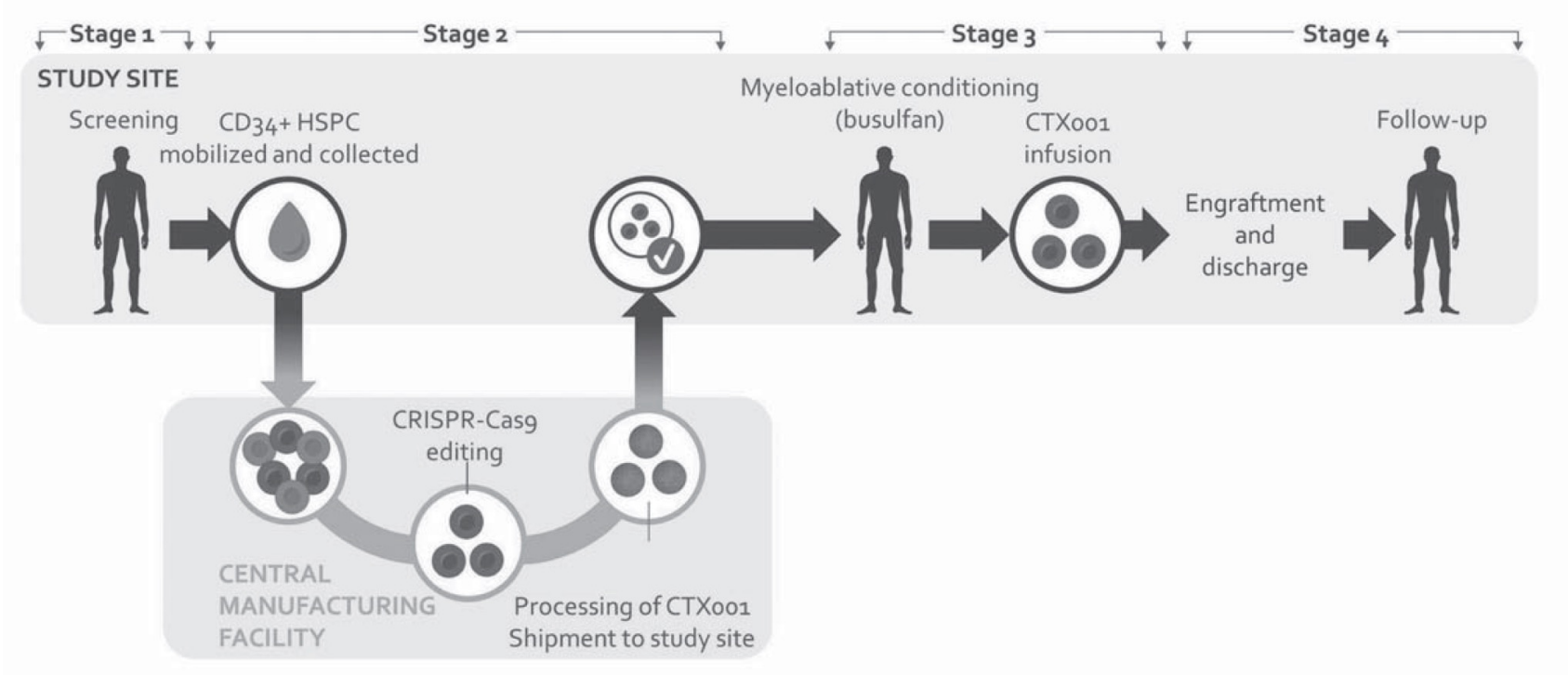


Motif that binds to goblin promotor



Casgevy – beta thalassemia / SCD treatment

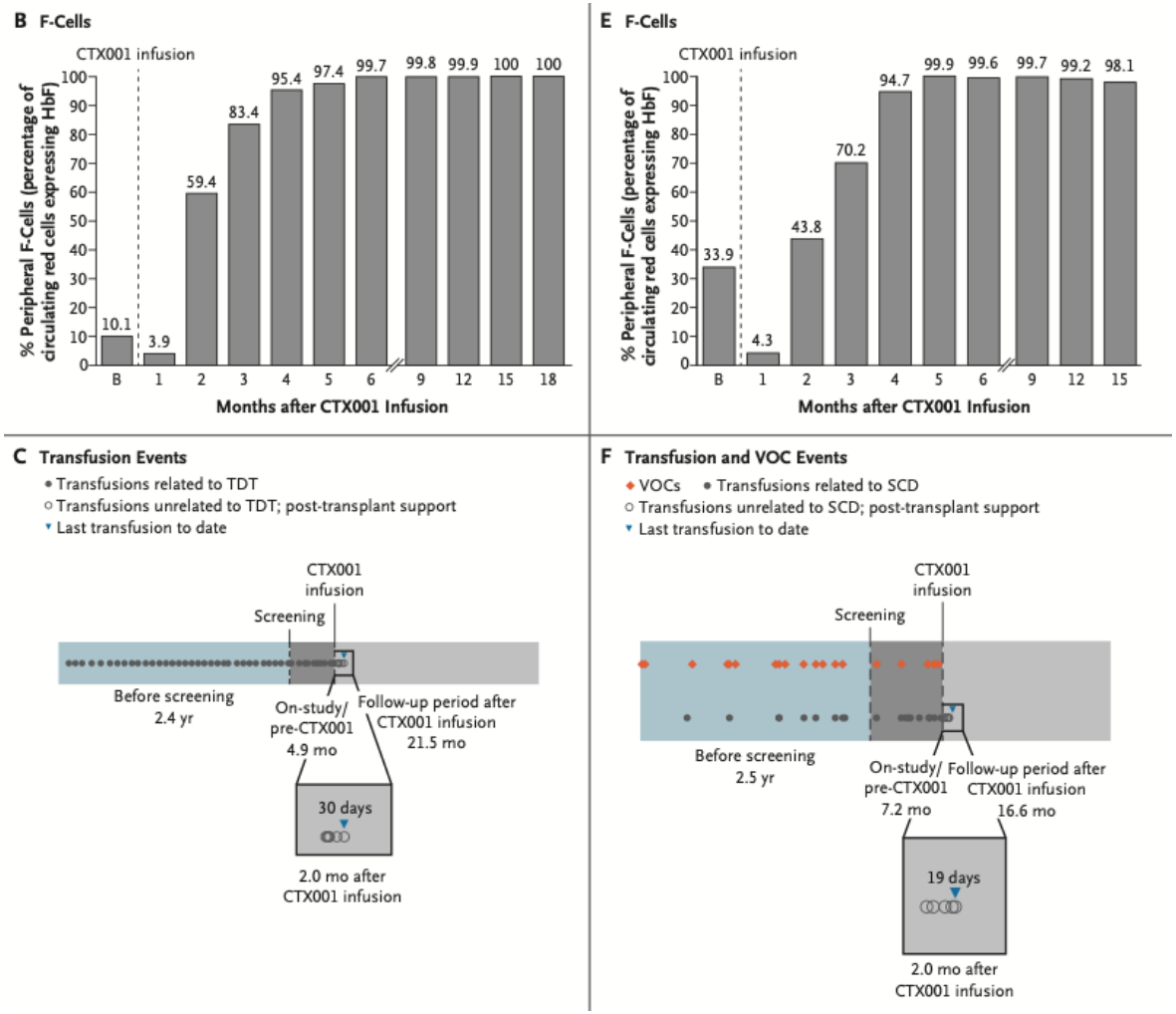
Casgevy treatment procedure



HSPC: hematopoietic stem and progenitor cells
CTX001: Casgevy

Casgevy – beta thalassemia / SCD treatment

Patient 1



Patient 2

Problems

Cost: 2.200.000 US$ (only casgevy)
Requires chemotherapy (side effects)
Off-target events
Lymphopenia
No long-term monitoring